$$\lambda^{-k} = \left(2-\sqrt{3}\right)^k = p_k - q_k\sqrt{3}\,. \tag{7}$$

1.5 **[1.0점]** $x_k = C\lambda^k$

λ

$$\lambda^{k+1} = 4\lambda^k - \lambda^{k-1}. \tag{8}$$

$$\lambda^2 - 4\lambda + 1 = 0\,, \tag{9}$$

$$\lambda_{1,2} = 2 \pm \sqrt{3}\,. \tag{10}$$

(3)

$$x_k = C_1\lambda_1^k + C_2\lambda_2^k\,, \tag{11}$$

C_1, C_2

$$\begin{aligned} x_0 = A \quad &\Rightarrow \quad C_1 + C_2 = A \\ x_0 = B \quad &\Rightarrow \quad C_1\lambda_1^N + C_2\lambda_2^N = B. \end{aligned} \tag{12}$$

$$\begin{cases} C_1 + C_2 = A, \\ C_1\lambda_1^N + C_2\lambda_2^N = B \end{cases} \Rightarrow \begin{cases} C_1 = \dfrac{B - A\lambda_2^N}{\lambda_1^N - \lambda_2^N}, \\ C_2 = \dfrac{A\lambda_1^N - B}{\lambda_1^N - \lambda_2^N}. \end{cases} \tag{13}$$

(11)

$$\begin{aligned} x_k = C_1\lambda_1^k + C_2\lambda_2^k = \frac{B - A\lambda_2^N}{\lambda_1^N - \lambda_2^N}\lambda_1^k + \frac{A\lambda_1^N - B}{\lambda_1^N - \lambda_2^N}\lambda_2^k = \\ = \frac{A\lambda_1^N\lambda_2^k - B\lambda_2^k + B\lambda_1^k - A\lambda_2^N\lambda_1^k}{\lambda_1^N - \lambda_2^N} = \frac{A\left(\lambda_1^{N-k} - \lambda_2^{N-k}\right) + B\left(\lambda_1^k - \lambda_2^k\right)}{\lambda_1^N - \lambda_2^N}. \end{aligned} \tag{14}$$

$\lambda_2 = \lambda_1^{-1}$.

1.6 **[0.5점]** $\lambda_{1,2}^k$

$$\lambda_1^k - \lambda_2^k = \lambda_1^k - \lambda_1^{-k} = \left(p_k + q_k\sqrt{3}\right) - \left(p_k - q_k\sqrt{3}\right) = 2q_k\sqrt{3}\,, \tag{15}$$

$$x_k = \frac{A\left(\lambda_1^{N-k} - \lambda_2^{N-k}\right) + B\left(\lambda_1^k - \lambda_2^k\right)}{\lambda_1^N - \lambda_2^N} = \frac{Aq_{N-k} + Bq_k}{q_N}\,. \tag{16}$$

2. 각기둥 모양의 철사틀 (7.0점)

2.1 **[0.8점]**

